

RADIUS

NDE SERVICES INC.

9004 – 107th Street
Fort St. John, BC V1J 5R5
(250) 787-0336 phone
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Client: Andritz Hydro Ltd

Date: 2012.05.28

Attention:

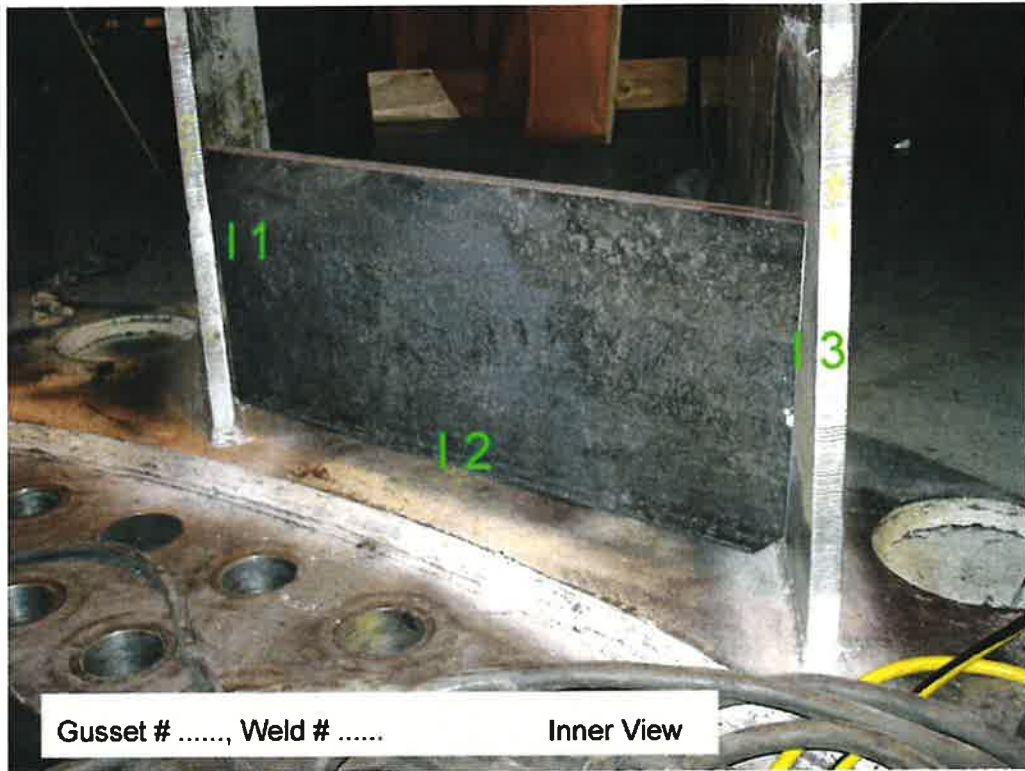
Page 1 of 1

Charge Code:

Ticket #:

Location: GM Shrum Generating Station

GUSSET WELD MAP





Installation and Commissioning
Quality Control Record

General Inspection Report

Q-0000-001

ITP step #: 6300-4

Revision # 00

Project: GMS

Unit : 4

Page 1 of 1

Equipment : Rotor Spider Painting Visual Inspection

As per comments in ITP 6300-4, new paint was to be applied over inspected areas.

However, BCH requested the inspected joints were left unpainted to facilitate any future weld inspection. Therefore, no new paint was applied on the recently welded joints.

These joints include the gussets, rotor rib welds, crossbraces welds, plug welds at entrance of each pocket and welded ring underneath the rotor spider.

NCR #: _____

Measurement Instruments	T°	Measured by: _____	dd-mmm-yyyy
Before: _____			
After: _____		ANDRITZ Rep.: <u>Jean-Luc Cote</u>	<u>20-Jun-2012</u>
		Cust. Repres. <u>[Signature]</u>	<u>Sept. 27, 2012</u>

General Inspection Report**Q-0000-001**

ITP step #: 6300-5 & 6

Revision # 00

Project: GMSUnit : G4

Page 1 of 1

Equipment : Rotor Rim offset and skew inspections

The measurements of the rotor rim skew and offset cannot be accurately measured from the dovetails due to large local variations. Measurements taken can vary for up to 1.5 mm from an adjacent lamination which prevents any data extrapolation. This condition appears to be existing from the initial stacking of the rotor laminations. See picture below for representation.



The only reliable measurement to provide offset and skew can be obtained by measuring the radial and tangential guides located inside the rotor rim. These measurements were recorded under ITP step 6000-6.



NCR #: _____

Measurement Instruments	T°	Measured by: _____	dd-mmm-yyyy
Before: _____			
After: _____		ANDRITZ Rep.: <u>Jean-Luc Cote</u>	<u>16-Jul-2012</u>
		Cust. Repres. _____	



GMS G4 - Rotor Rim Circularity and Concentricity June 21st

ITP 6000-19 & 6300-10

Measured by: Christian Maheux

Faro Laser Tracker X02000501577 (Calibrated March 29th 2012)

Summary of GMS G4 rotor rim measurements (in inch)

	Before Shrink	After Shrink	Tolerance
Circ Bottom	0.033	0.023	0.061
Circ Middle	0.025	0.027	0.061
Circ Top	0.033	0.026	0.061
Circ Total	0.023	0.024	0.046

Conc bottom	0.009	0.006	0.009
Conc Middle	0.010	0.005	0.009
Conc Top	0.011	0.003	0.009
Conc Total	0.010	0.005	0.009

Verticality	0.043	0.041	0.046
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GMS G4 - Rotor Rim Circularity and Concentricity After Shrink June 21st 2012

ITP 6300-10

Faro Laser Tracker X02000501577 (Calibrated March 29th 2012)

Measured by Christian Maheux

Values in mm

	Bottom Rim, lower	Bottom Rim, upper	avg bottom	Middle Rim, Lower	Middle rim, Upper	avg middle	Top Rim, Lower	Top Rim, upper	Avg top	Total average
1	5231.30	5231.14	5231.22	5231.07	5231.22	5231.15	5231.22	5232.02	5231.62	5231.29
2	5231.41	5231.27	5231.34	5231.02	5231.23	5231.13	5231.12	5231.97	5231.55	5231.31
3	5231.39	5231.29	5231.34	5231.15	5231.19	5231.17	5231.10	5232.01	5231.56	5231.33
4	5231.43	5231.33	5231.38	5231.28	5231.30	5231.29	5231.15	5231.80	5231.48	5231.37
5	5231.35	5231.24	5231.30	5231.35	5231.15	5231.25	5231.20	5231.45	5231.33	5231.29
6	5230.85	5231.08	5230.97	5231.13	5231.16	5231.15	5231.03	5231.20	5231.12	5231.07
7	5230.79	5231.02	5230.91	5231.01	5231.15	5231.08	5230.97	5231.10	5231.04	5231.00
8	5231.02	5231.10	5231.06	5231.04	5231.02	5231.03	5231.03	5231.44	5231.24	5231.09
9	5231.10	5231.01	5231.06	5231.16	5231.25	5231.21	5231.12	5231.64	5231.38	5231.19
10	5231.24	5231.30	5231.27	5231.32	5231.23	5231.28	5231.19	5231.65	5231.42	5231.31
11	5231.14	5231.36	5231.25	5231.27	5231.22	5231.25	5231.13	5231.65	5231.39	5231.28
12	5231.08	5231.41	5231.25	5231.23	5231.20	5231.22	5231.06	5231.59	5231.33	5231.25
13	5230.67	5231.19	5230.93	5231.13	5231.11	5231.12	5230.82	5231.30	5231.06	5231.03
14	5230.73	5231.04	5230.89	5231.05	5231.16	5231.11	5230.92	5231.30	5231.11	5231.02
15	5231.04	5231.19	5231.12	5231.12	5231.23	5231.18	5231.05	5231.71	5231.38	5231.20
16	5230.97	5231.07	5231.02	5231.18	5231.24	5231.21	5231.20	5231.93	5231.57	5231.23
17	5231.07	5231.18	5231.13	5231.24	5231.34	5231.29	5231.25	5231.80	5231.53	5231.29
18	5230.91	5231.12	5231.02	5231.18	5231.27	5231.23	5231.23	5231.88	5231.56	5231.23
19	5231.02	5231.00	5231.01	5230.99	5231.20	5231.10	5231.04	5231.93	5231.49	5231.16
20	5230.82	5230.63	5230.73	5230.68	5230.98	5230.83	5230.75	5231.78	5231.27	5230.90
21	5230.84	5230.52	5230.68	5230.52	5230.88	5230.70	5230.71	5231.79	5231.25	5230.83
22	5231.16	5230.61	5230.89	5230.58	5230.85	5230.72	5230.87	5231.82	5231.35	5230.94
23	5230.92	5230.59	5230.76	5230.46	5230.93	5230.70	5230.63	5231.64	5231.14	5230.83
24	5230.85	5230.68	5230.77	5230.69	5231.02	5230.86	5230.84	5231.60	5231.22	5230.91
25	5230.97	5230.99	5230.98	5231.02	5231.19	5231.11	5231.01	5231.68	5231.35	5231.12
26	5230.68	5230.92	5230.80	5231.03	5231.11	5231.07	5231.02	5231.53	5231.28	5231.02
27	5230.62	5230.88	5230.75	5230.98	5231.08	5231.03	5230.97	5231.55	5231.26	5230.98
28	5230.76	5230.88	5230.82	5230.97	5231.08	5231.03	5230.95	5231.72	5231.34	5231.03
29	5230.90	5231.03	5230.97	5230.87	5230.99	5230.93	5230.91	5231.83	5231.37	5231.05
30	5230.56	5230.75	5230.66	5230.74	5230.84	5230.79	5230.76	5231.60	5231.18	5230.84
31	5230.68	5230.83	5230.76	5230.84	5230.85	5230.85	5230.76	5231.65	5231.21	5230.90
32	5230.97	5231.02	5231.00	5231.04	5230.96	5231.00	5231.07	5231.74	5231.41	5231.10
33	5230.95	5231.25	5231.10	5231.22	5231.08	5231.15	5231.13	5231.75	5231.44	5231.20
34	5231.01	5231.23	5231.12	5231.30	5231.22	5231.26	5231.14	5231.74	5231.44	5231.25
35	5230.81	5231.14	5230.98	5231.05	5231.11	5231.08	5230.96	5231.43	5231.20	5231.07
36	5231.01	5230.96	5230.99	5230.87	5230.91	5230.89	5230.77	5231.32	5231.05	5230.96
37	5230.53	5230.63	5230.58	5230.58	5230.67	5230.63	5230.63	5231.13	5230.88	5230.67
38	5230.64	5230.65	5230.65	5230.47	5230.71	5230.59	5230.57	5231.32	5230.95	5230.70
39	5231.14	5230.94	5231.04	5230.76	5230.86	5230.81	5230.67	5231.70	5231.19	5230.99
40	5231.22	5231.18	5231.20	5231.04	5230.92	5230.98	5230.86	5231.81	5231.34	5231.15
41	5231.08	5231.28	5231.18	5231.13	5231.16	5231.15	5231.00	5231.82	5231.41	5231.22
42	5230.88	5231.12	5231.00	5231.15	5231.16	5231.16	5231.01	5231.54	5231.28	5231.13
43	5230.94	5231.14	5231.04	5231.08	5231.14	5231.11	5231.02	5231.51	5231.27	5231.12
44	5230.58	5230.94	5230.76	5230.94	5230.89	5230.92	5230.88	5231.30	5231.09	5230.90
45	5230.62	5230.95	5230.79	5230.87	5230.88	5230.88	5230.86	5231.32	5231.09	5230.90
46	5231.00	5231.01	5231.01	5230.90	5230.99	5230.95	5231.00	5231.56	5231.28	5231.05
47	5230.81	5230.97	5230.89	5230.89	5230.94	5230.92	5230.87	5231.52	5231.20	5230.98
48	5231.02	5231.03	5231.03	5230.85	5230.96	5230.91	5231.01	5231.78	5231.40	5231.07



GMS G4 - Rotor Rim Circularity and Concentricity Prior to Shrink June 8th 2012

ITP 6009-19

Faro Laser Tracker X02000501577 (Calibrated March 29th 2012)

Measured by Christian Maheux

Values in mm

	Bottom Rim, lower	Bottom Rim, upper	avg bottom	Middle Rim, Lower	Middle rim, Upper	avg middle	Top Rim, Lower	Top Rim, upper	Avg top	Total average
1	5229.90	5229.95	5229.92	5229.88	5230.17	5230.02	5230.07	5230.56	5230.31	5230.06
2	5229.91	5229.88	5229.90	5229.87	5230.16	5230.02	5229.94	5230.39	5230.16	5230.01
3	5229.85	5230.08	5229.97	5229.87	5230.07	5229.97	5229.87	5230.29	5230.08	5230.00
4	5230.11	5230.09	5230.10	5230.09	5230.08	5230.08	5229.90	5230.31	5230.10	5230.09
5	5230.23	5230.19	5230.21	5230.20	5230.11	5230.16	5230.06	5230.21	5230.14	5230.17
6	5229.94	5230.19	5230.07	5230.17	5230.15	5230.16	5229.92	5230.07	5229.99	5230.08
7	5229.93	5230.16	5230.05	5230.05	5230.12	5230.09	5229.86	5229.95	5229.91	5230.03
8	5230.08	5230.18	5230.13	5230.10	5230.22	5230.16	5229.97	5230.15	5230.06	5230.12
9	5230.24	5230.27	5230.26	5230.33	5230.43	5230.38	5230.19	5230.31	5230.25	5230.30
10	5230.59	5230.73	5230.66	5230.66	5230.61	5230.63	5230.46	5230.57	5230.51	5230.61
11	5230.62	5230.93	5230.77	5230.79	5230.73	5230.76	5230.56	5230.63	5230.60	5230.73
12	5229.92	5230.49	5230.21	5230.60	5230.63	5230.61	5230.52	5230.71	5230.61	5230.46
13	5229.81	5230.57	5230.19	5230.60	5230.67	5230.64	5230.58	5230.81	5230.69	5230.49
14	5229.94	5230.51	5230.23	5230.64	5230.79	5230.72	5230.60	5230.88	5230.74	5230.54
15	5230.19	5230.65	5230.42	5230.66	5230.87	5230.76	5230.71	5231.08	5230.90	5230.67
16	5230.18	5230.71	5230.44	5230.74	5230.84	5230.79	5230.78	5231.17	5230.98	5230.71
17	5230.13	5230.56	5230.35	5230.66	5230.85	5230.76	5230.83	5231.10	5230.96	5230.65
18	5230.15	5230.60	5230.37	5230.65	5230.74	5230.69	5230.71	5231.08	5230.90	5230.62
19	5230.28	5230.51	5230.40	5230.51	5230.75	5230.63	5230.67	5231.05	5230.86	5230.60
20	5230.18	5230.15	5230.16	5230.28	5230.65	5230.47	5230.48	5231.05	5230.77	5230.43
21	5230.31	5230.07	5230.19	5230.17	5230.59	5230.38	5230.45	5231.00	5230.73	5230.40
22	5230.41	5230.20	5230.30	5230.25	5230.54	5230.39	5230.54	5231.02	5230.78	5230.46
23	5230.39	5230.36	5230.38	5230.35	5230.67	5230.51	5230.51	5230.85	5230.68	5230.50
24	5230.48	5230.61	5230.54	5230.60	5230.94	5230.77	5230.68	5231.02	5230.85	5230.70
25	5230.37	5230.78	5230.58	5230.76	5230.88	5230.82	5230.75	5230.96	5230.85	5230.74
26	5230.28	5230.72	5230.50	5230.81	5230.83	5230.82	5230.73	5230.97	5230.85	5230.71
27	5230.32	5230.76	5230.54	5230.72	5230.85	5230.79	5230.58	5230.93	5230.75	5230.69
28	5230.45	5230.70	5230.57	5230.73	5230.74	5230.73	5230.61	5230.96	5230.78	5230.69
29	5230.48	5230.75	5230.61	5230.63	5230.72	5230.68	5230.54	5230.94	5230.74	5230.67
30	5230.22	5230.54	5230.38	5230.51	5230.56	5230.54	5230.39	5230.80	5230.59	5230.49
31	5230.20	5230.55	5230.38	5230.43	5230.47	5230.45	5230.40	5230.75	5230.57	5230.45
32	5230.31	5230.54	5230.42	5230.51	5230.48	5230.49	5230.47	5230.69	5230.58	5230.49
33	5230.38	5230.65	5230.52	5230.61	5230.54	5230.58	5230.55	5230.66	5230.61	5230.56
34	5230.32	5230.71	5230.52	5230.65	5230.56	5230.60	5230.53	5230.67	5230.60	5230.57
35	5230.10	5230.61	5230.35	5230.47	5230.51	5230.49	5230.41	5230.42	5230.42	5230.42
36	5229.76	5230.41	5230.09	5230.31	5230.38	5230.34	5230.29	5230.38	5230.34	5230.25
37	5229.70	5230.15	5229.93	5230.13	5230.29	5230.21	5230.14	5230.33	5230.24	5230.11
38	5229.82	5230.09	5229.96	5230.02	5230.07	5230.05	5230.02	5230.48	5230.25	5230.06
39	5230.09	5230.13	5230.11	5230.01	5230.24	5230.13	5230.00	5230.53	5230.26	5230.15
40	5230.05	5230.10	5230.08	5230.09	5230.25	5230.17	5230.02	5230.52	5230.27	5230.16
41	5230.07	5230.30	5230.18	5230.32	5230.48	5230.40	5230.38	5230.80	5230.59	5230.36
42	5230.04	5230.48	5230.26	5230.46	5230.63	5230.55	5230.46	5230.82	5230.64	5230.46
43	5230.04	5230.50	5230.27	5230.48	5230.55	5230.51	5230.52	5230.78	5230.65	5230.46
44	5229.77	5230.32	5230.05	5230.43	5230.57	5230.50	5230.50	5230.79	5230.65	5230.37
45	5229.70	5230.32	5230.01	5230.34	5230.59	5230.46	5230.48	5230.79	5230.63	5230.34
46	5229.88	5230.21	5230.05	5230.19	5230.57	5230.38	5230.41	5230.77	5230.59	5230.31
47	5229.78	5230.00	5229.89	5229.95	5230.23	5230.09	5230.22	5230.70	5230.46	5230.11
48	5229.77	5229.88	5229.82	5229.87	5230.13	5230.00	5230.16	5230.74	5230.45	5230.04

Installation and Commissioning Quality Control Record

Welding visual inspection

Q-0400-011

ITP step #: 6300-11

Revision # 00

Project: GMS

Unit : 4

Page 1 of 1

Applicable standard

- ☒ CSA W59 Article 5.9
☐ ANSI/AWS D1.1 Article 3.6
☐ ASME B31.1 article 136.4
 (piping)

Welding joint inspected	ITP step #	Any superficial cracks	Any lack of fusion between welding and the base metal	Any crater	Welding profiles acceptable according to standard	Dimension in conformity with drawing	Date (dd-mmm-yyyy)	Initials
Rotor pole stop blocks	6300-11	No	No	No	Yes	Yes	30-Aug-2012	JLC



Comments : The old stop blocks were removed by grinding. The new adjustable stop blocks were welded (6 mm), vertical down position of each side of the blocks and overhead position for the bottom of the block. The elevations was kept as per old blocks as directed by BC Hydro.

NCR #: _____

Measurement Instruments	T°	Measured by: _____	dd-mmm-yyyy
Visual	Before: _____	ANDRITZ Rep.: <u>Jean-Luc Cote</u>	<u>30-Aug-2012</u>
	After: _____	Cust. Repres. <u>[Signature]</u>	<u>Sept. 27, 2012</u>

Installation and Commissioning Quality Control Record

General Inspection Report

Q-0000-001

 ITP step #: 6300-12

 Revision # 00

 Project: GMS

 Unit : 4

Page 1 of 1

 Equipment : Visual on rotor cleaning

The rotor poles were cleaned prior to completing the final electrical testing Monday september 23rd.



The final rotor spider and rim cleaning was completed september 26th. All surfaces were wiped down with rags and simple green.



NCR #: _____

Measurement Instruments	T°	Measured by:	dd-mmm-yyyy
Visual	Before: _____	ANDRITZ Rep.: <u>Jean-Luc Cote</u>	<u>27-Sep-2012</u>
	After: _____	Cust. Repres. <u>[Signature]</u>	<u>Sept 27, 2012</u>

Installation and Commissioning
Quality Control Record**Rotor Poles Hipot & Megger Test****Q-0500-002**ITP step #: 6300-13Project: GMS Generating StationUnit : 4Revision # 00Page 1 of 2**SAFETY CHECK LIST**

Equipment Tested: Rotor Pole Insulation Test voltage: 1000 ☐ Volt AC
Test Leader: Gavin Frie ☒ Volt DC
Equipment operator: Gavin Frie
Power ON checked by: Gavin Frie
Power OFF checked by: Gavin Frie
Watchman: Geordie Barr Location: Service bay floor
First Aid Coverage by: Geordie Barr

Components to be grounded: Rotor polesDistance of approach: 3 FeetTheoretical grounding duration time: 1 minuteGrounding done: ☒ Yes ☐ No ☐ N/AInsulated rubber gloves available: ☒ Yes ☐ No ☐ N/ABarrier tape in place: ☒ Yes ☐ No ☐ N/ALOTO Locks available: ☐ Yes ☐ No ☒ N/AGround stick is checked and ready: ☒ Yes ☐ No ☐ N/ARequired PPE is available and used: ☒ Yes ☐ No ☐ N/A

Additional checks and comments:

Test recorded on new rotor poles while they are located on powerhouse floor prior to
assembly.

NCR #: _____

Measurement Instruments	T°	Measured by: <u>Gavin Frie</u>	<u>01-Aug-12</u>
<u>Megger Tester</u>	Before: <u>21 C</u>	<u>Geordie Barr</u>	<u>1-Aug-2012</u>
<u>417360</u>	After: <u>21 C</u>	ANDRITZ Rep.: <u>Jean-Luc Cote</u>	<u>3-Aug-2012</u>
		Cust. Repres. _____	_____

Installation and Commissioning Quality Control Record

Rotor Poles Hipot & Megger Test

Q-0500-002

ITP step #: 6300-13

Revision # 00

Project: GMS Generating Station

Unit : 4

Page 2 of 2

Number of readings: 48

Readings in : G Ohms

Pole	Megger test	Temperature	Corrected at 40 C	Pole	Megger test	Temperature	Corrected at 40 C
1	71 G Ohms	21 C	19.17	25	69 G Ohms	21 C	18.63
2	55 G Ohms	21 C	14.85	26	72 G Ohms	21 C	19.44
3	80 G Ohms	21 C	21.60	27	77 G Ohms	21 C	20.79
4	78 G Ohms	21 C	21.06	28	68 G Ohms	21 C	18.36
5	72 G Ohms	21 C	19.44	29	75 G Ohms	21 C	20.25
6	74 G Ohms	21 C	19.98	30	74 G Ohms	23 C	22.94
7	69 G Ohms	21 C	18.63	31	70 G Ohms	21 C	18.90
8	73 G Ohms	21 C	19.71	32	73 G Ohms	21 C	19.71
9	72 G Ohms	21 C	19.44	33	78 G Ohms	21 C	21.06
10	76 G Ohms	21 C	20.52	34	60 G Ohms	21 C	16.20
11	76 G Ohms	21 C	20.52	35	68 G Ohms	21 C	18.36
12	69 G Ohms	21 C	18.63	36	92 G Ohms	23 C	28.52
13	106 G Ohms	21 C	28.62	37	80 G Ohms	21 C	21.60
14	78 G Ohms	21 C	21.06	38	76 G Ohms	21 C	20.52
15	74 G Ohms	21 C	19.98	39	76 G Ohms	21 C	20.52
16	73 G Ohms	21 C	19.71	40	70 G Ohms	21 C	18.90
17	67 G Ohms	21 C	18.09	41	74 G Ohms	21 C	19.98
18	57 G Ohms	21 C	15.39	42	78 G Ohms	21 C	21.06
19	68 G Ohms	21 C	18.36	43	84 G Ohms	21 C	22.68
20	68 G Ohms	21 C	18.36	44	64 G Ohms	21 C	17.28
21	69 G Ohms	21 C	18.63	45	76 G Ohms	21 C	20.52
22	86 G Ohms	21 C	23.22	46	64 G Ohms	21 C	17.28
23	68 G Ohms	21 C	18.36	47	62 G Ohms	21 C	16.74
24	85 G Ohms	21 C	22.95	48	68 G Ohms	21 C	18.36

Notes: 1- Megger test at 1000 V

Acceptance criteria is higher than 10 Gigaohms.
corrected to 40 C

NCR #: _____

*** Pole #30 & #36 were re-taken August 1st

Measurement Instruments	T°	Measured by:	Gavin Frie	01-Aug-12
Megger Tester	Before: 21 C		Geordie Barr	1-Aug-2012
417360	After: 21 C	ANDRITZ Rep.:	Jean-Luc Cote	3-Aug-2012
		Cust. Repres.		

Installation and Commissioning
Quality Control Record**General Inspection Report****Q-0000-001**ITP step #: 6300-14Revision # 00Project: GMSUnit : 4Page 1 of 2Equipment : High Frequency Impedence Testing of poles (420 Hz) before assembly of poles

Pole #	Amps	Volts	Watts	% Power Factor	Coil Temp C
1	3.169	10.215	7.952	24.55	19.1
2	3.273	10.243	8.190	24.40	22.8
3	3.221	10.264	7.650	23.17	23.3
4	3.251	10.204	7.280	21.96	22.2
5	3.241	10.236	8.082	24.37	23.2
6	3.392	10.283	8.499	24.32	23.3
7	3.200	10.245	8.026	24.40	18.4
8	3.200	10.253	7.390	22.51	20.2
9	3.210	10.234	8.086	24.61	24.5
10	3.248	10.188	7.930	23.95	22.6
11	3.270	10.203	7.946	23.81	23.9
12	3.192	10.236	7.928	24.27	25.3
13	3.255	10.234	8.262	24.91	23.3
14	3.220	10.299	7.820	23.53	19.8
15	3.219	10.221	7.329	22.27	19.1
16	3.211	10.263	8.460	25.66	24.5
17	3.222	10.249	8.380	25.38	23.3
18	3.250	10.211	7.957	23.97	23.1
19	3.261	10.230	8.920	26.44	23.1
20	3.220	10.226	7.829	23.78	19.3
21	3.222	10.221	8.329	25.29	23.8
22	3.258	10.228	7.215	21.66	27.8
23	3.314	10.266	8.566	25.18	19.3
24	3.230	10.252	7.336	22.15	23.1

***Test Completed using BCH testing equipment

NCR #: _____

Measurement Instruments	T°	Measured by:	Jared Szabo	dd-mmm-yyyy
F2350 Test System 143	Before: <u>22 C</u>		G. Frie, G. Barr	Aug-3-2012
Megger Power Multimeter	After: <u>22 C</u>	ANDRITZ Rep.:	Jean-Luc Cote	7-Aug-2012
301		Cust. Repres.		



Installation and Commissioning
Quality Control Record

General Inspection Report

Q-0000-001

ITP step #: 6300-14

Revision # 00

Project: GMS

Unit : 4

Page 2 of 2

Equipment : High Frequency Impedence Testing of poles (420 Hz) before assembly of poles

Pole #	Amps	Volts	Watts	% Power Factor	Coil Temp C
25	3.252	10.268	8.456	25.45	23.6
26	3.190	10.242	7.480	22.90	19.3
27	3.160	10.261	8.440	26.04	23.1
28	3.200	10.224	7.349	22.44	24.6
29	3.187	10.292	8.315	25.34	19.0
30	3.179	10.196	7.946	24.51	19.3
31	3.170	10.245	7.165	22.07	24.5
32	3.210	10.232	8.101	24.66	20.0
33	3.220	10.274	8.129	24.56	21.2
34	3.250	10.257	8.179	24.53	22.1
35	3.232	10.239	7.251	21.92	23.2
36	3.198	10.242	7.809	23.84	21.2
37	3.219	10.233	8.263	25.09	20.1
38	3.252	10.308	7.944	23.72	24.1
39	3.250	10.285	7.616	22.97	19.9
40	3.250	10.214	7.682	23.14	19.7
41	3.221	10.293	7.478	22.56	21.6
42	3.252	10.140	8.572	25.51	21.2
43	3.210	10.253	8.218	24.94	20.3
44	3.241	10.236	8.056	24.28	23.7
45	3.324	10.247	8.172	23.99	22.8
46	3.180	10.239	7.760	23.84	25.1
47	3.283	10.188	7.676	22.95	23.3
48	3.221	10.272	7.682	26.11	23.3

***Test Completed using BCH testing equipment

NCR #: _____

Measurement Instruments	T°	Measured by: <u>Jared Szabo</u>	dd-mmm-yyyy
<u>F2350 Test System 143</u>	Before: <u>22 C</u>	<u>G. Frie, G. Barr</u>	<u>Aug-3-2012</u>
<u>Megger Power Multimeter</u>	After: <u>22 C</u>	ANDRITZ Rep.: <u>Jean-Luc Cote</u>	<u>7-Aug-2012</u>
<u>301</u>		Cust. Repres. _____	



Installation and Commissioning
Quality Control Record

General Inspection Report

Q-0000-001

ITP step #: 6300-15 &16

Revision # 00

Project: GMS

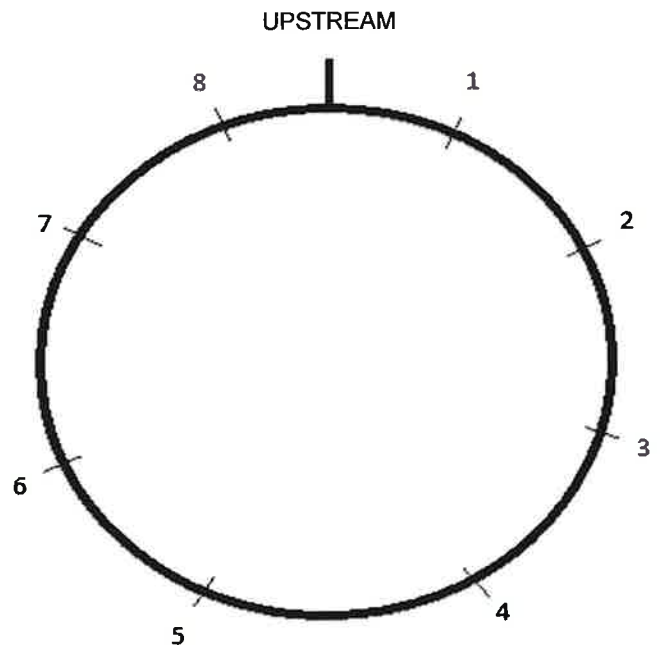
Unit : 4

Page 1 of 3

Equipment : Rotor Level at reassembly

Rotor is sitting onto pedestal on service bay, all poles are installed however pole elevations have not been set

Point	Measurement (inch)
1	0.275
2	0.273
3	0.272
4	0.272
5	0.271
6	0.272
7	0.273
8	0.276



NCR #: _____

Measurement Instruments	T°	Measured by:	Mike Rados	dd-mmm-yyyy
Optical Level N3	Before: <u>24.6 C</u>		Mike Kelly	13-Aug-2012
#TL 20435	After: <u>24.6 C</u>	ANDRITZ Rep.:	Jean-Luc Cote	14-Aug-2012
		Cust. Repres.	<u>[Signature]</u>	<u>Sept 27, 12</u>

Installation and Commissioning Quality Control Record

General Inspection Report

Q-0000-001

ITP step #: 6300-15 & 16

Revision # 00

Project: GMS

Unit : 4

Page 2 of 3

Equipment : Rotor Pole elevation at reassembly (values in inch)

Pole #	Dovetail length / 2	N3 bottom dovetail	N3 bottom coupling	N3 difference needed	N3 difference obtained	Delta
1	39.000	11.134	23.844	12.698	12.710	-0.012
2*	38.965	11.334	23.844	12.662	12.510	0.152
3	38.984	11.149	23.844	12.682	12.695	-0.013
4*	38.992	11.224	23.844	12.690	12.620	0.070
5	38.984	11.179	23.844	12.682	12.665	0.017
6	39.000	11.146	23.844	12.698	12.698	0.000
7	38.984	11.174	23.844	12.682	12.670	0.012
8	38.965	11.174	23.844	12.662	12.670	-0.008
9	38.983	11.174	23.844	12.681	12.670	0.011
10	39.030	11.114	23.844	12.727	12.730	-0.003
11	38.965	11.174	23.844	12.662	12.670	-0.008
12	39.000	11.139	23.844	12.698	12.705	-0.007
13*	38.980	11.170	23.845	12.678	12.675	0.003
14*	38.970	11.110	23.820	12.668	12.710	-0.042
15	38.969	11.015	23.715	12.666	12.700	-0.034
16	38.965	11.065	23.715	12.662	12.650	0.012
17	38.970	11.060	23.715	12.668	12.655	0.013
18	39.000	11.025	23.715	12.698	12.690	0.008
19*	38.965	11.115	23.715	12.662	12.600	0.062
20	39.016	11.010	23.715	12.713	12.705	0.008
21	38.970	11.042	23.715	12.668	12.673	-0.005
22	38.970	11.050	23.715	12.668	12.665	0.003
23	38.984	11.025	23.715	12.682	12.690	-0.008
24	38.970	11.060	23.715	12.668	12.655	0.013

*Target distance between magnetic center is 26.302" (provided by BCH in FAM-BCH-AH-081)

**Positive value = higher than 26.302", Negative value = lower than 26.302"

***Poles #2,4,19, 29, 30, 33, 34, 36, 40, 41, 42, 43, 46 cannot be adjusted lower due to stop block location

***Poles #13, 14, 39, 44 were originally out-of-tolerance, stop blocks were relocated on August 22nd as per BC Hydro request

NCR #: _____

Measurement Instruments	T°	Measured by:	dd-mmm-yyyy
Optical Level N3	Before: <u>24.6 C</u>	<u>Mike Rados</u>	<u>22-Aug-2012</u>
#TL 20435	After: <u>24.6 C</u>	<u>Mike Kelly</u>	<u>22-Aug-2012</u>
		ANDRITZ Rep.: <u>Jean-Luc Cote</u>	<u>22-Aug-2012</u>
		Cust. Repres: <u>[Signature]</u>	<u>Sept 27, 12</u>

Installation and Commissioning Quality Control Record

General Inspection Report

Q-0000-001

ITP step #: 6300-15 &16

Revision # 00

Project: GMS

Unit : 4

Page 3 of 3

Equipment : Rotor Pole elevation at reassembly (values in inch)

Pole #	Dovetail length / 2	N3 bottom dovetail	N3 bottom coupling	N3 difference needed	N3 difference obtained	Delta
25	38.965	11.075	23.715	12.662	12.640	0.022
26	38.970	11.040	23.715	12.668	12.675	-0.007
27	39.012	11.028	23.715	12.709	12.687	0.022
28	38.984	11.005	23.715	12.682	12.710	-0.028
29*	39.004	11.090	23.715	12.702	12.625	0.077
30*	38.965	11.095	23.715	12.662	12.620	0.042
31	38.980	11.040	23.715	12.678	12.675	0.003
32	39.020	10.975	23.715	12.717	12.740	-0.023
33*	38.984	11.085	23.715	12.682	12.630	0.052
34*	38.965	11.175	23.715	12.662	12.540	0.122
35	38.965	11.055	23.715	12.662	12.660	0.002
36*	38.970	11.105	23.715	12.668	12.610	0.058
37	39.000	11.055	23.715	12.698	12.660	0.038
38	38.980	11.037	23.715	12.678	12.678	0.000
39*	39.000	11.135	23.840	12.698	12.705	-0.007
40*	38.980	11.125	23.715	12.678	12.590	0.088
41*	38.970	11.155	23.715	12.668	12.560	0.108
42*	38.990	11.165	23.715	12.688	12.550	0.138
43*	39.000	11.120	23.715	12.698	12.595	0.103
44*	38.984	11.155	23.840	12.682	12.685	-0.003
45	38.965	11.060	23.715	12.662	12.655	0.007
46*	38.969	11.115	23.715	12.666	12.600	0.066
47	38.965	11.060	23.715	12.662	12.655	0.007
48	38.969	11.054	23.715	12.666	12.661	0.005

*Target distance between pole magnetic center and rotor coupling flange is 26.302" (provided by BCH in FAM-BCH-AH-081)

**Positive value = higher than 26.302", Negative value = lower than 26.302"

***Poles #2,4,19, 29, 30, 33, 34, 36, 40, 41, 42, 43, 46 cannot be adjusted lower due to stop block location

***Poles #13, 14, 39, 44 were originally out-of-tolerance, stop blocks were relocated on August 22nd as per BC Hydro request

NCR #: _____

Measurement Instruments	T°	Measured by:	Mike Rados	dd-mmm-yyyy
Optical Level N3	Before: 24.6 C		Mike Kelly	22-Aug-2012
#TL 20435	After: 24.6 C	ANDRITZ Rep.:	Jean-Luc Cote	22-Aug-2012
		Cust. Repres.		